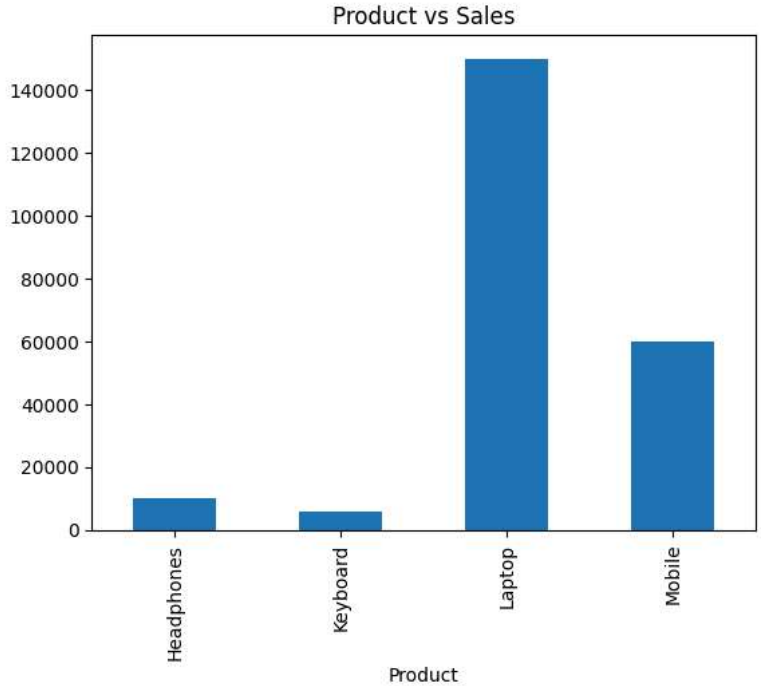
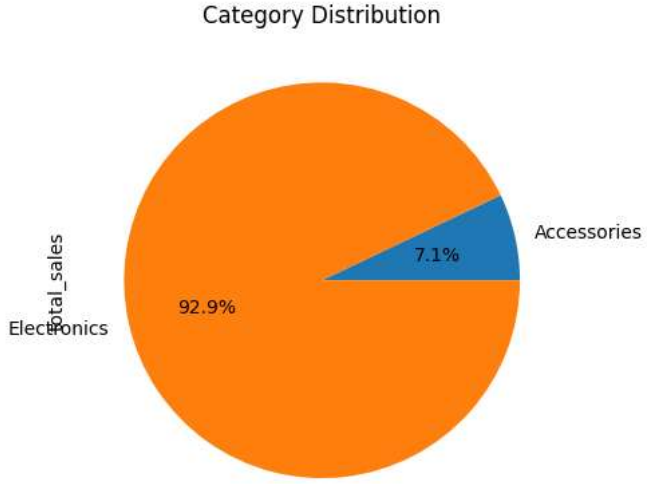
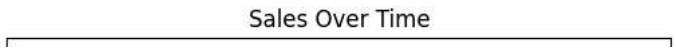
```
# 1. Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# 2. Load Data
df = pd.read_csv('/content/sales.csv')

# 3. Data Preview
print(df.head())

# 4. Feature Engineering
df['total_sales'] = df['Quantity'] * df['Price']

# 5. Analysis
print("Total Revenue:", df['total_sales'].sum())

print("\nSales by Product:")
print(df.groupby('Product')['total_sales'].sum())

print("\nSales by City:")
print(df.groupby('City')['total_sales'].sum())

# 6. Visualization
df.groupby('Product')['total_sales'].sum().plot(kind='bar')
plt.title("Product vs Sales")
plt.show()
```

```
   order_id order_date    City     Product     Category  Quantity  Price
0      1001  1/10/2025    Pune      Laptop  Electronics         2  50000
1      1002  1/11/2025  Mumbai      Mobile  Electronics         3  20000
2      1003  1/12/2025    Pune  Headphones  Accessories         5   2000
3      1004  1/13/2025  Nagpur    Keyboard  Accessories         4   1500
4      1005  1/14/2025  Mumbai      Laptop  Electronics         1  50000
Total Revenue: 226000

Sales by Product:
Product
Headphones     10000
Keyboard        6000
Laptop        150000
Mobile         60000
Name: total_sales, dtype: int64

Sales by City:
City
Mumbai    110000
Nagpur      6000
Pune      110000
Name: total_sales, dtype: int64
```




```
df.groupby('Category')['total_sales'].sum().plot(kind='pie', autopct='%1.1f%%')
plt.title("Category Distribution")
plt.show()
```



```
df['order_date'] = pd.to_datetime(df['order_date'])
df.groupby('order_date')['total_sales'].sum().plot(kind='line')
plt.title("Sales Over Time")
plt.show()
```